







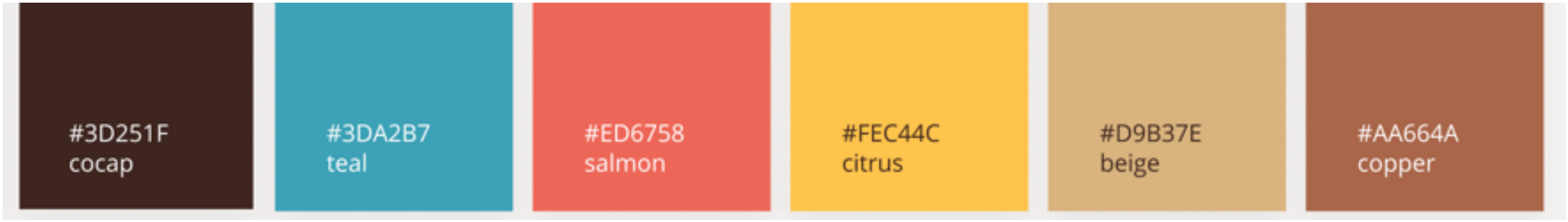


$\alpha$



$\beta$





#3D251F  
cocap

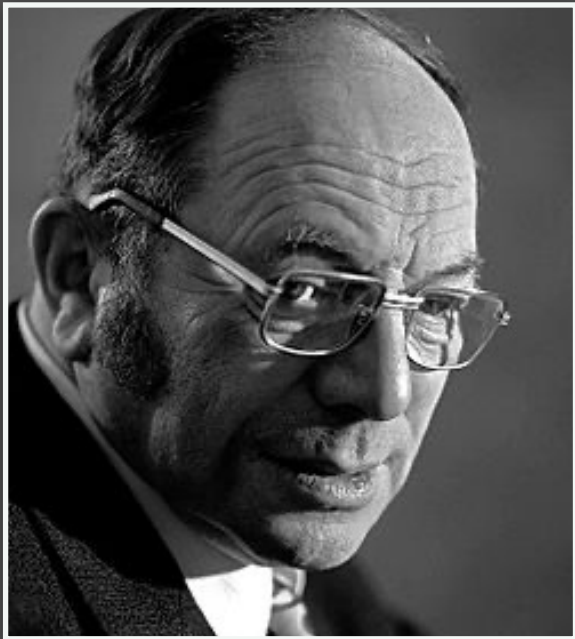
#3DA2B7  
teal

#ED6758  
salmon

#FEC44C  
citrus

#D9B37E  
beige

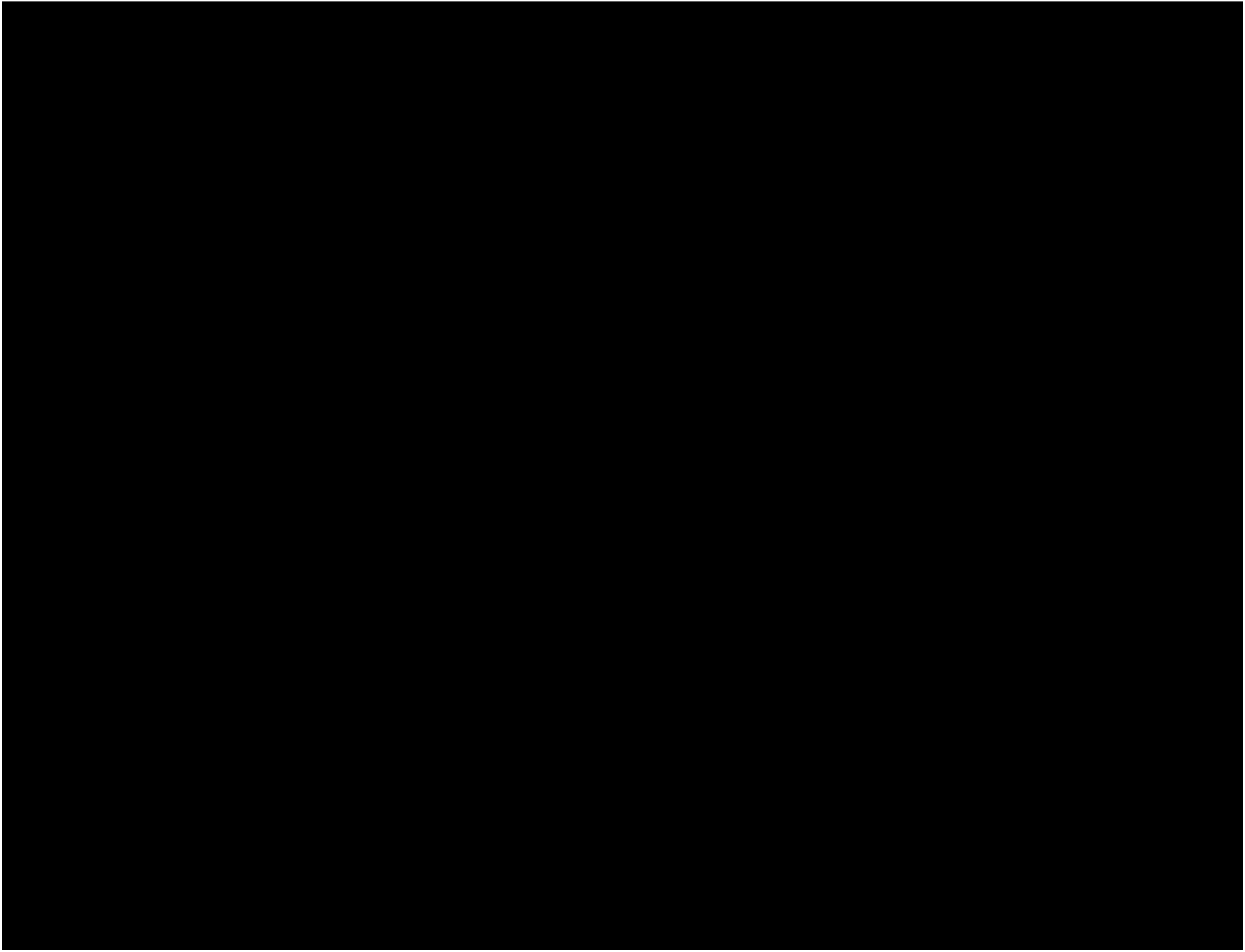
#AA664A  
copper



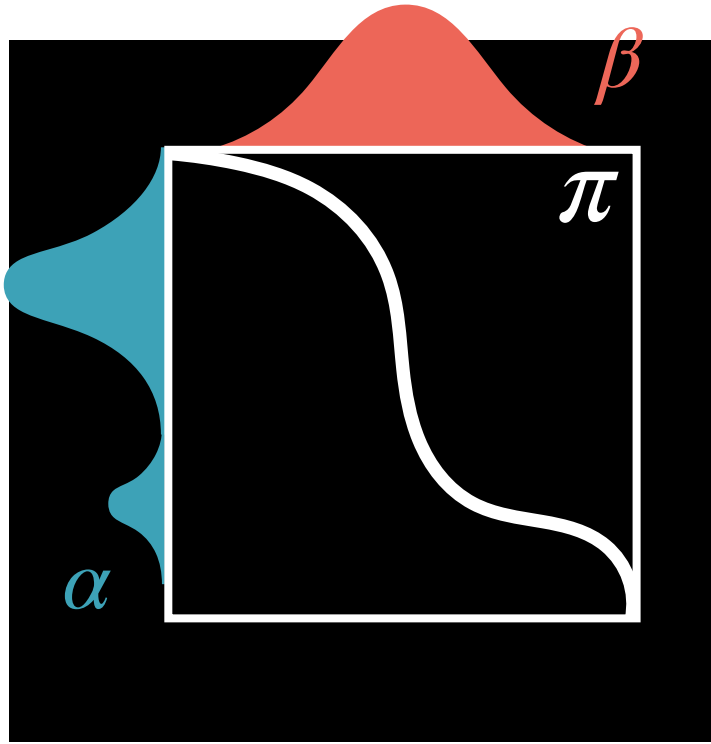
# Kantorovich's Problem:

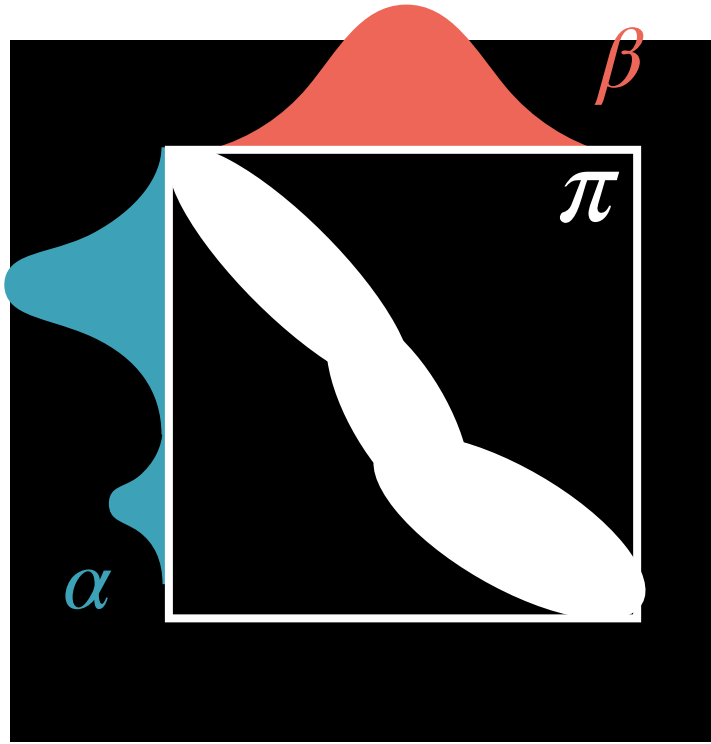
(Continuous version)

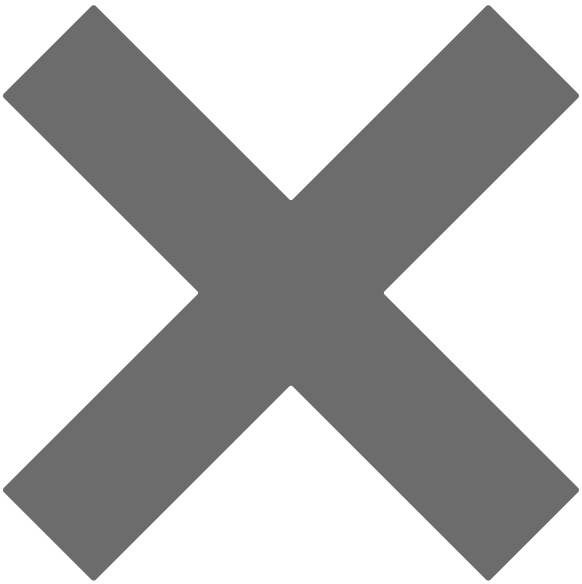
$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$







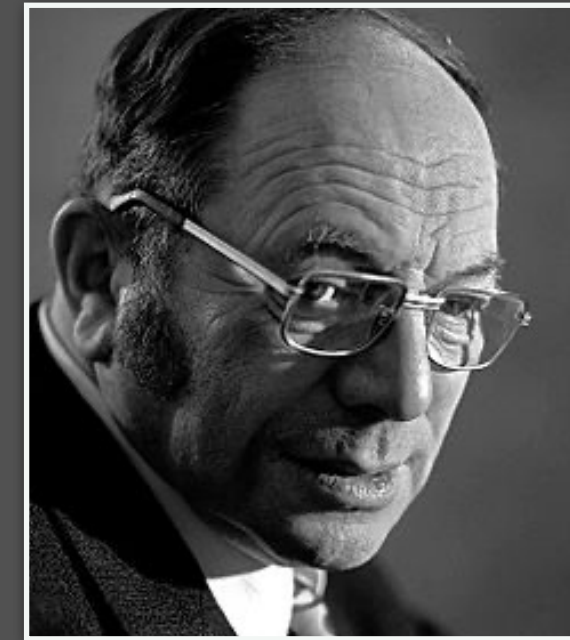




# Brenier's Theorem

## Assumptions:

1.  $c(x, y) = \|x - y\|^2$
2.  $\alpha$  has a density  $\rho_\alpha = \frac{d\alpha}{dx}$



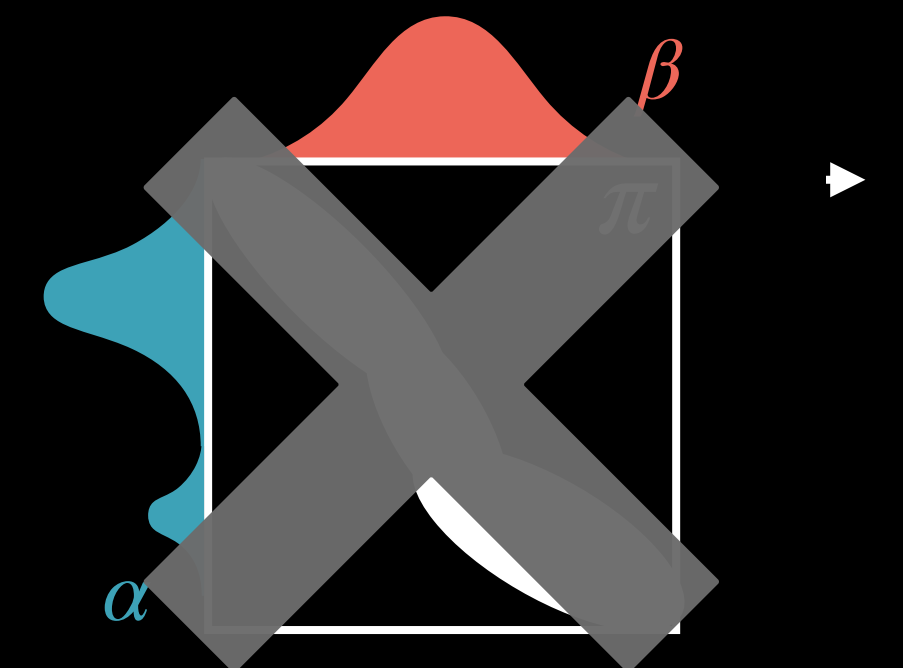
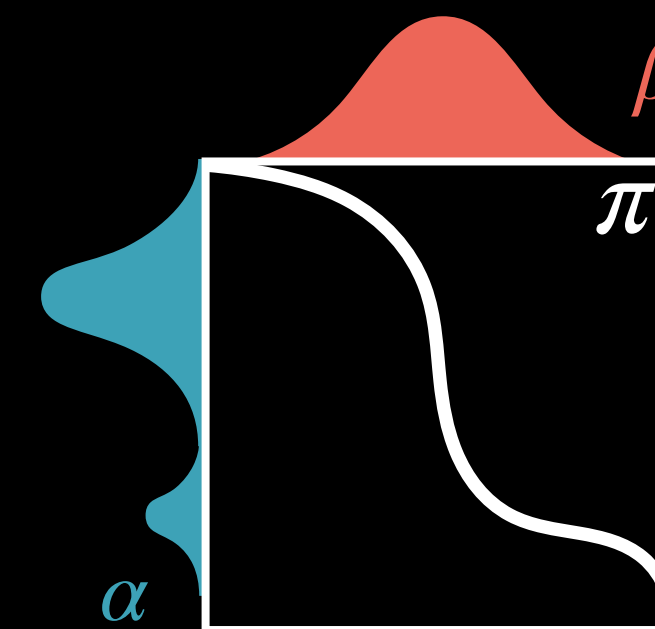
## Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

## Theorem [Brenier, 1991]:

1. There exists a **unique** coupling  $\pi$  that solves the Kantorovich problem
2. The coupling is deterministic, supported on the graph of a Monge map:  $\pi = (\text{Id}, T)_\# \alpha$
3.  $T$  is the gradient of a convex function  
Formally: the unique  $T = \nabla \varphi$  such that  $\varphi$  is **convex** and  $(\nabla \varphi)_\# \alpha = \beta$



# Brenier's Theorem: Significance

- Provides conditions under which Monge and Kantorovich **problems are equivalent**
- In such cases, Monge map has very special structure ( $\nabla$  of a convex function)
- This structure is exploited by recent neural OT methods